

Maxwill Lin

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Google Code Jam Top 0.56% & ETH Global Paris \$9k Winner. First-principles problem solver who builds core functions at leading companies across diverse fields: GenAI, ads ranking, autonomous systems, theoretical cryptography and quantitative trading.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. in Computer Science (Atlanta Main Campus); GPA: 4.0/4.0

08/2023 – 12/2024

- Teaching Assistant of Blockchain and Cryptocurrencies (24 spring) and Machine Learning (24 fall).
- **Relevant coursework:** [Advanced ML](#), [DL for Text](#), [Advanced Algorithms and Uncertainty](#), [Efficient ML](#), Programming Languages Design, [Deep Reinforcement Learning](#), Brain-inspired ML, Information Security
- Examples: prompting LLMs to write codes to play games — fine-tuning, quantization, and pruning transformers — mathematics foundations in energy-based models, reinforcement learning, online learning, and online algorithms.

National Yang Ming Chiao Tung University

Hsinchu, Taiwan

B.S. in Computer Science, B.S. in Applied Mathematics; GPA: 3.8/4.0

09/2018 – 06/2021

- Top 3 university in Taiwan. Presidential Award (top 5%). A+ in [advanced algorithms](#), advanced linear algebra, etc.
- **Relevant coursework:** theoretical computer science, mathematics, machine learning, computer systems

EXPERIENCE

Vmax AI

San Francisco, CA

Founding Member of Technical Staff; Python, PyTorch, slime, Harbor, Ray, RL Infrastructure

03/2026 - present

- First hire; designed and built the core training system end-to-end solo: RL infrastructure orchestrating hundreds of GPUs, a self-improving task generator, and a rollout framework running thousands of concurrent sandboxes.
- Achieved a +50% relative improvement over an open-source base model on SWE-bench Pro under a constrained context window using synthetic tasks.
- Contributed open-source changes used by the training stack: added [CISPO advantage estimator](#) to slime; added [dspy.RLM agent](#) to Harbor with sandbox tool bridge and tests; fixed Harbor [agent installs](#) and [adapter docs](#).

Meta

Sunnyvale, CA

Research Engineer, MRS; Python, PyTorch, Dataswarm, Internal ML frameworks

02/2025 - 03/2026

- Meta Recommendation System (MRS) aims to unify RecSys across Meta products using state-of-the-art techniques.
- Trained billion-parameter autoregressive models for user-ad interaction; achieved 0.07% NE gain for mobile feed CTR prediction; featured in org-wide EngEx shoutout as 0→1 usage of a new internal ML framework.
- Investigated LLMs for CTR prediction; built scalable evaluation pipeline reducing runtime from hours to minutes; observed zero-shot performance scaled with model size, reasoning effort, and in-context data quantity/quality; post-training on ads data improved AUC from 0.51 to 0.61.
- Led cross-functional design of offline RL dataset for long-term user value optimization; drove alignment across data engineers, scientists, and RL researchers
- #1 code contributor in org of 40 engineers as a new hire; promoted to next level within 9 months; 156 diffs, 19.6K SLOC, 77 code reviews; tooling fix and wiki contributions reach 400+ unique engineers.

Tesla Autopilot

Palo Alto, CA

Autopilot Software Engineering Intern; C++, Python, C, Bazel, Jinja2, Git

05/2024 - 08/2024

- Developed scripts to automatically refactor codebase to avoid unnecessary dependencies and reduce related CI by ~50%.
- Co-designed and built a high-performance C API for inter-process communication to decouple ML models from core Autopilot systems, accelerating iteration cycles; enabled declarative API usage by building a C++ reflection library, partially open-sourced at [fixed-containers](#).
- Optimized module build time by code-gen independent translation units, resulting in a ~50× speed up.

BTQ

Taipei, Taiwan

Research Engineer; Rust, Python, Post Quantum Cryptography, Zero-knowledge Proofs

08/2022 - 08/2023

- Identified and fixed a critical vulnerability in a proposed signature aggregation protocol, preventing signature forgery
- Designed, implemented, and optimized a signature aggregation protocol in Rust, achieving 400× verification time speedup. The protocol encodes the Falcon signature verifier as templated arithmetic graphs and verifies them recursively w/ Plonky2.
- Researched cryptographic primitives (zkSNARKs, PCD, recursion, folding), focusing on both mathematical constructions and code implementations

Kronos Research

Taipei, Taiwan

Quantitative Research Intern; C++, Bash, Python, Machine Learning

06/2022 - 08/2022

- Selected with the best assessment score from hundreds of applicants to a top-tier crypto prop trading firm.
- Developed an order tracking tool to quantify execution slippages

- Designed and implemented an alpha signal that improves R2 score by 42.4% when combined with existing signals built on similar data source.

Sudo Research Labs

Taipei, Taiwan

Research and Development Intern; EVM, Solidity, Blockchains, consensus algorithms

03/2022 - 06/2022

- Identified risks of blockchain/DeFi protocols by reviewing the code bases and prevented millions of investment losses
- Researched consensus/layer 2/cross-chain protocols and contributed to open-source documentation of Arbitrum (largest layer 2 protocol with 2.6 B total value locked), [Polygon Bor](#), and [decentralized thoughts](#).

Institute of Information Science, Academia Sinica

Taipei, Taiwan

Research Intern; Python, PyTorch, AllenNLP, Graph Neural Networks

07/2020 - 09/2020

- Applied GNNs and cross attention to natural language inference by considering it as dependency graph matching. Implemented the model modularly to be compatible with the [AllenNLP](#) framework.

AWARDS AND HONORS

Google Code Jam 2020

ranked 541st in round 3 (**top 0.56%**)

ETH Global Paris 2023

won 7 prizes and 8863 USD in total out of 1400 attendees and 321 projects.

Admitted to Jianguo High School Science Class

STEM-gifted class in top 1 high school in Taiwan

ALGORITHMS AND MACHINE LEARNING PROJECTS

- **LLM Post-training Datasets:** Built large-scale LLM post-training datasets for crossword/mathematical problem solving using Ray and Google BigQuery.
- **Distributed SWE-Bench:** Developed a lightweight, concurrent SWE-Bench eval library that runs in 15 minutes on a cluster of sandboxes, providing fine-grained error reports and an interactive dashboard for filtering and analysis. Came with modern task management using Taskfile, Pytest, precommit hooks, and CI/CD (artifacts export, E2E NOOP tests), with strongly typed, functional-styled, and modularized code. Within a month.
- **Coursework: Efficient Machine Learning:** Implemented various efficient machine learning techniques: including finetuning and pruning with early-bird tickets hypothesis, dynamic quantization of LLMs, distributed training, etc.
- **Final Project: Language Agents for Games:** Prompted language models to write code to play games and reflect based on interaction with other agents.
- **Final Project: Where did it go wrong? LLM Agent with Error Locator:** Experimented a novel prompt engineering technique inspired by policy gradient.
- **Presentation: Attention Sinks:** Presentation of a method to do infinite length inference of language model efficiently.
- **Labs Project: EazyNoGo:** Programmed a NoGo AI based on Monte Carlo Tree Search with Rapid Action Value Estimation in C++. This is the individual study project in the 2019 fall semester at [NCTU CGI lab](#) advised by [Dr.I-Chen Wu](#).
- **Writing Samples: Theoretical Aspects of Modern Cryptography:** The links for the [syllabus](#) and [coursework](#)
- **Labs Project: Best Chain Protocols:** A presentation for a group meeting covering the setting, analysis, and scaling solutions of best chain protocols
- **Writing Samples: Exponential Arithmetic Speedup by Quantum Walks:** The links for the [final report](#) and its [presentation](#)
- **Labs Project: Solution to k -MaxVD for Trees in $O(kn)$:** Designed an algorithm to calculate the maximum number of vertices that a k subset can dominate in a tree with dynamic programming and optimization. Individual study project in junior year advised by [Dr. Chiuyuan Chen](#) and co-advised by [Dr. Meng-Tsung Tsai](#).
- **Writing Samples: Treap Tutorial for NCTU PCCA winter camp:** [NCTU PCCA](#) is the competitive programming club in NCTU, the material is for the winter camp in 2020

PUBLICATIONS

Maxwill Lin, Meng-Tse Wu, and Keh-Yih Su. Syntax-aware Natural Language Inference with Graph Matching Networks. 2020 International Conference on Technologies and Applications of Artificial Intelligence. IEEE, 2020.

SKILLS AND TECHNOLOGIES

Languages: English (**GRE: 332/340, TOEFL: 108/120**), Chinese (native)

Programming Languages: C++, Python, Rust, LaTeX, Solidity, TypeScript, JavaScript

Technologies: PyTorch, Ray, slime, Harbor, SGLang, Docker, git, AWS, SQL, ethers.js, Hardhat, Plonky2